

Lerato Emmanuel Mgwangqa

JUNIOR SOFTWARE AND DATA ENGINEER | EXPERTISE IN MERN STACK, AI/ML, DATA MODELING, AND NETWORK ARCHITECTURE

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Professional Summary

Data Engineering Graduate Trainee and BSc graduate in Mathematical and Computer Sciences, currently pursuing BSc Honours in Computer Science. Strong foundation in data engineering, analytics, cloud platforms, and backend systems, with hands-on experience supporting enterprise data pipelines, governance, and reporting. Experienced with Databricks Lakehouse concepts, Azure data services, ETL workflows, and Agile delivery practices. Research-oriented with exposure to explainable AI, graph-based modeling, and security analytics.

Experience

Sasol | Data Engineering Trainee

2025-Present

- Support enterprise data platforms through **ETL processing, data transformation, and storage management**
- Assist with **data governance, compliance, and data quality validation** processes
- Perform SQL-based data analysis and reporting for business stakeholders
- Participating in **Agile team ceremonies** including sprint planning, stand-ups, and retrospectives
- Contribute to audit readiness through data integrity checks and documentation
- Collaborate with cross-functional teams on data tools, pipelines, and operational improvements

Technologies & Practices: SQL, ETL Pipelines, Data Quality, Agile/Scrum, Reporting, Enterprise Data Systems

Sol Plaatje University | Peer Mentor

2024-2024

- Provide academic and personal support to over 20+ first time entering students
- Enhanced their academic performance and study skills.
- Organize study groups and workshops, contributing to a collaborative learning environment
- Collaborate with faculty to develop action plans for struggling students

ALX Program | Software Engineering Trainee, Online

2023-2024

- Languages & Frameworks: JavaScript (ES6+), TypeScript, Node.js, Express.js, Python, & C
- Backend Development: Advanced MySQL, NoSQL, Redis, Queuing systems, caching, pagination, user authentication, and data security

- DevOps & Systems Engineering: Linux basics, web servers (Nginx), application servers, load balancing, monitoring, CI/CD, and web stack debugging
- Completed all this as comprehensive program focused on modern backend development techniques and tools.

EDUCATION

Sol Plaatje University, BSc Hons Computer Sciences

2025 – Present (*Part-time*)

Major: Software Engineering | Minor: AI, Machine Learning & Security

Honours Research: Explainable Phishing Detection and Authentication Risk Modeling Using Metadata Link Graphs and Threat Intelligence

Specialized Experience: Applied ML, graph modeling (NetworkX), explainable AI, anomaly detection, phishing detection research

Sol Plaatje University, BSc Mathematical and Computer Sciences

2022-2024

Major: Computer Sciences and Statistics | Minor: Mathematics and Applied Mathematics

SKILLS & ABILITIES

Programming & Databases

- Python, C/C++ (OOP), R, Shell Scripting
- JavaScript (ES6+), TypeScript, Node.js, Express
- SQL, MySQL, MongoDB (NoSQL), Redis

Machine Learning & Explainable AI

- Pandas, NumPy, Scikit-learn
- Graph-based modeling (NetworkX)
- Explainable AI (SHAP, LIME)
- Predictive modeling and anomaly detection

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Data Visualization & BI

- Power BI
- Excel
- Matplotlib, Seaborn

Data Engineering & Analytics

- End-to-end data pipeline design
- ETL and data transformation workflows
- Data quality checks, validation, and reconciliation
- Transactional vs analytical workloads (OLTP vs OLAP)
- Schema design, normalization, and denormalization
- Metadata-driven processing and feature engineering

Databricks & Lakehouse Architecture

- Databricks workspace fundamentals
- Databricks Jobs, Tasks, and Workflows
- Interactive clusters vs Job clusters
- Medallion Architecture (Bronze, Silver, Gold)
- Batch vs streaming processing concepts
- Trigger-based and scheduled job execution

Cloud & Microsoft Tools:

- Microsoft Azure fundamentals
- Azure Data Factory (ETL orchestration concepts)
- Azure Synapse Analytics (analytics concepts)
- Azure Cosmos DB APIs (real-time ingestion concepts)
- Microsoft Fabric fundamentals
- Cloud cost awareness and security basics

Agile, DevOps & DataOps

- Agile & Scrum delivery practices
- Sprint planning, stand-ups, reviews, retrospectives
- CI/CD concepts and Azure DevOps exposure
- Git-based version control workflows
- Environment separation (dev/test/prod)
- Monitoring, logging, and pipeline debugging

Soft Skills: Technical Writing, Collaboration, Time Management, Public Speaking, Analytical Thinking, Resilience, Excellent computer skills, Software Development Cycle

PROJECTS & PRACTICAL EXPERIENCE

GitHub Top Repo Analysis

- **Description:** Analyzed GitHub's top repositories to identify trends in programming languages and technologies.
- **Tools Used:** Python, Streamlit, GitHub API, Data Analysis Techniques

Residence-Based Management System

- **Description:** Developed a system to streamline and manage student residence data, improving administrative efficiency.
- **Tools Used:** MySQL, HTML5, CSS3, JavaScript

C Printf Implementation

- **Description:** Created a simplified version of the C printf function to deepen understanding of C's low-level functionality.
- **Tools Used:** C Programming Language

Chatbot Project

- **Description:** Built a chatbot for intent classification and NLP, using machine learning for enhanced user interaction.
- **Tools Used:** Python, Flask, Scikit-Learn, NLTK, Pandas, HTML, CSS

E-Learning Platform for Skill Development Learnaz Hub

- **Description:** An engaging e-learning platform offering courses across various fields, designed to make learning accessible for all ages.
- **Tools Used:** React, TailwindCSS, Node.js, Express, MongoDB, Git, GitHub.

Data Science Workflow Practice (GitHub: practical1) | 2025

- Built a full ML pipeline: **data preprocessing, visualization, feature engineering, and model evaluation.**
- Applied Logistic Regression on heart disease dataset, achieving strong classification metrics (accuracy, precision, recall, F1).
- Tools: **Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter** to deliver end-to-end insights.

Supervised Machine Learning Practice (GitHub: ml_prac2) | 2025

- Built and evaluated ML models (Decision Tree, Random Forest, Logistic Regression, KNN, Naïve Bayes) on datasets like Iris, Bank, and Diabetes.
- Applied preprocessing, visualization, and evaluation; achieved up to **100% accuracy** (Random Forest on Iris).
- Tools: **Python, Scikit-learn, Pandas, Matplotlib, Jupyter.**

Certifications

ALX Software Engineering Program (Graduated: August 2024) - [Certificate Awarded](#)

Peer Mentor , Sol Plaatje University - [Peer Mentor Certificate 2024 - Lerato Mgwangqa](#)

Microsoft Azure Fundamentals (AZ-900)

Microsoft Azure Data Fundamentals (DP-900)

References

“Available Upon Request”